Global and Local Projection-Based Reduced-Order Models

2D Burgers and Hypersonic Vehicle Campaigns

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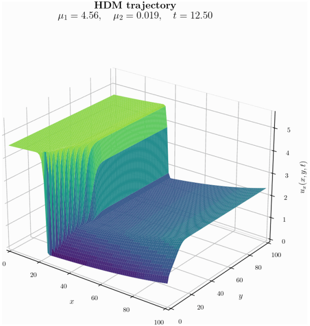
August 5, 2026

2D Burgers Campaign



Global and local HPROM validation

Linear POD bases, quadratic manifolds, and GPR manifolds

Setup for Paper Campaign

Two-dimensional parametric inviscid Burgers problem

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} + \frac{\partial \mathbf{G}(\mathbf{U})}{\partial y} = \mathbf{S}(x;\boldsymbol{\mu}), \qquad \mathbf{U} = \begin{bmatrix} u_x \\ u_y \end{bmatrix}.$$

$$\mathbf{S}(x;\boldsymbol{\mu}) = \begin{bmatrix} 0.02\exp(\mu_2 x) \\ 0 \end{bmatrix}, \qquad u_x(0,y,t;\boldsymbol{\mu}) = \mu_1.$$

μ_1: inflow value $\quad$ μ_2: exponential source variation $\quad$ $\boldsymbol{\mu} \in [4.25, 5.50] \times [0.015, 0.030]$

Test points:

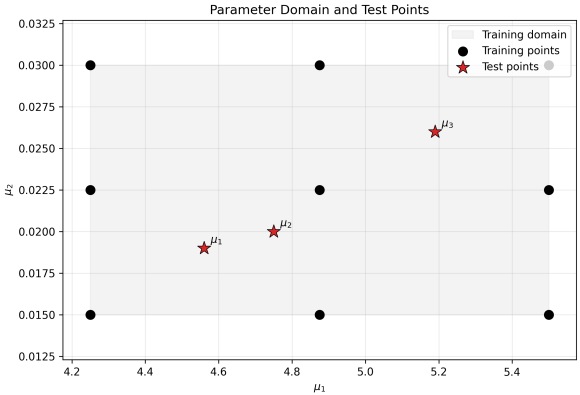
$$\boldsymbol{\mu}_1 = (4.56, 0.019), \quad \boldsymbol{\mu}_2 = (4.75, 0.020), \quad \boldsymbol{\mu}_3 = (5.19, 0.026).$$

HDM: $\Omega = [0,100]^2$, 250×250 finite-volume cells.

Methods included:

Global HPROM	Local HPROM
Global HQPROM	Local HQPROM
Global HPROM–GPR	Local HPROM–GPR

Parameter Domain and Test Points



Online Performance (250×250): Avg/Max Error over 3 Test Points

Computational model	n (N for HDM)	$\bar{n}$	N_e	avg $\mathbb{RE}_{2,\mathbf{u}}$ (%)	max $\mathbb{RE}_{2,\mathbf{u}}$ (%)	Wall-clock time (min)	Speedup
HDM	125 000	–	62 500	–	–	12.512	–
HPROM	96	–	4 824	1.030	1.096	0.743	16.841
HQPROM	39	–	3 505	0.871	0.933	0.888	14.084
HPROM-GPR	20	131	1 801	0.691	0.845	0.371	33.698
Local HPROM ($N_c = 3$)	35–48	–	3 649	0.827	0.854	0.377	33.168
Local HQPROM ($N_c = 3$)	11–15	–	1 164	0.716	0.844	0.311	40.259
Local HPROM-GPR ($N_c = 3$)	9	60–97	811	0.713	0.789	0.282	44.379

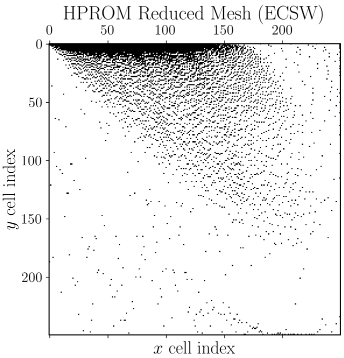
Speedup is computed with respect to the HDM runtime, $t_{\mathrm{HDM}} = 750.702$ s.

Online timings were measured on Sherlock using one node, 24 MPI ranks, and 24 cores.

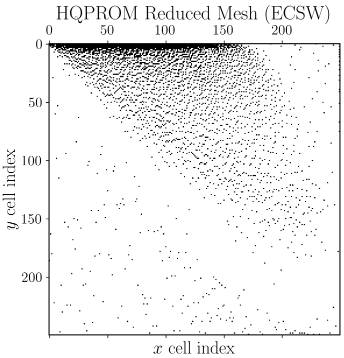
N_e denotes the number of sampled ECSW elements ($N_e = 62{,}500$ for HDM/full mesh).

Local ROM results shown here use $N_c = 3$ clusters; the $N_c = 10$ results are reproduced in the appendix.

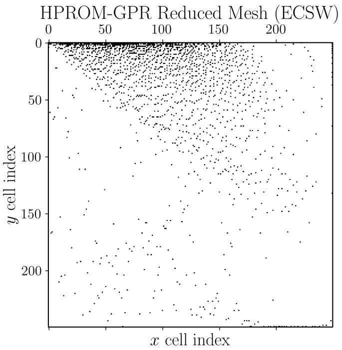
Global Reduced Meshes (ECSW)



HPROM (N_e = 4,824)

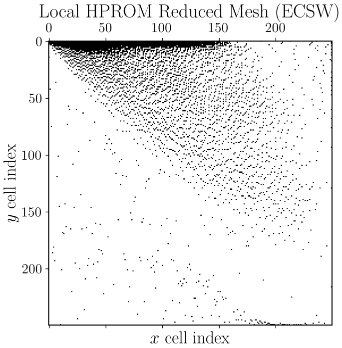


HQPROM (N_e = 3,505)

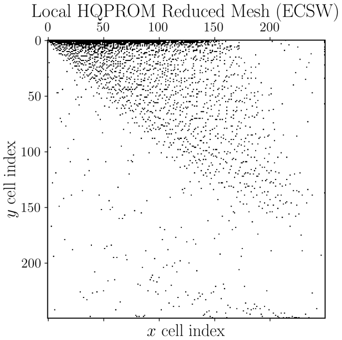


HPROM-GPR (N_e = 1,801)

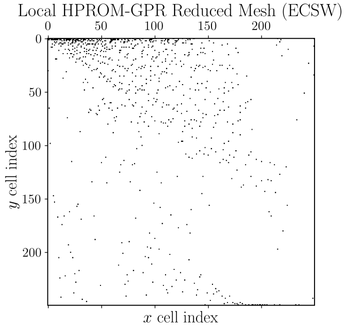
Local Reduced Meshes (ECSW), $N_c = 3$



Local HPROM (N_e = 3,649)



Local HQPROM (N_e = 1,164)

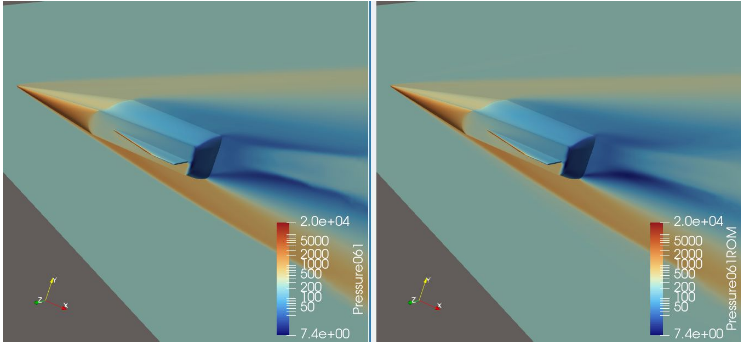


Local HPROM-GPR (N_e = 811)

Burgers Conclusions

- All local ROMs improve runtime performance relative to their global counterparts.
- Local HPROM–GPR is the fastest model: 0.282 min and $44.379\times$ speedup.
- Its speed comes primarily from the smaller ECSW mesh, with 811 selected elements versus 1,801 for global HPROM–GPR. Local GPR evaluation is also cheaper because the active-cluster regressor is trained with fewer data than the global regressor.
- Reported local-model runtimes include cluster selection and any cluster-transfer cost. The decision and transfer rules are formulation-specific and are defined in the appendix.

Hypersonic Vehicle Campaign

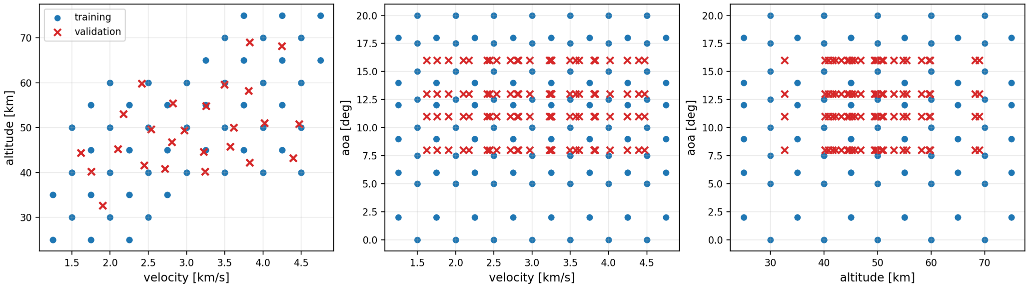


Global and local HPROM validation

Linear POD bases and RBF manifolds

Training and Validation Space

363 HDM training cases; 100 independent HDM validation cases; four local clusters with 20% overlap; ECSW tolerance 10^{-3}.



HGV2 ROM Configurations

- Global linear HPROM uses one POD basis. Local linear HPROM uses four POD bases and assigns each online state to its nearest cluster.
- Global HPROM–RBF uses $n = 8$ primary coordinates and predicts the remaining coordinates with one RBF map.
- Local HPROM–RBF uses one $n = 8$ RBF map per cluster. The local primary and secondary dimensions are reported in the next table.
- Every POD basis and RBF map is trained from the same 363 HDM parameter cases. ECSW is an independent offline construction step.

HGV2 Online Performance and Validation Accuracy

Coefficient errors are global relative L^2 values over 100 independent validation cases. Online times are means over 10 validation cases on Sherlock: one node, 24 MPI ranks, and 24 cores.

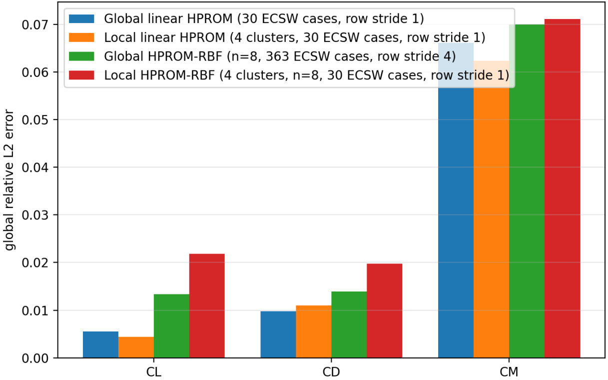
Model	n	$\bar{n}$	C_D [%]	C_L [%]	C_M [%]	**Online time [s]**	**Speedup**
Global linear HPROM	362	–	**0.975**	0.554	6.620	24.03	1.00
Local linear HPROM, $N_c = 4$	95–168	–	1.101	**0.444**	**6.242**	6.80	3.53
Global HPROM–RBF	8	354	1.388	1.335	7.004	6.00	4.00
Local HPROM–RBF, $N_c = 4$	8	117–140	1.978	2.183	7.117	**2.28**	**10.54**

For linear HPROM, n is the full POD rank. For RBF, n and $\bar{n}$ are the primary and secondary dimensions; local $\bar{n}$ values are ranges across the four cluster bases. Speedup is relative to global linear HPROM.

Online time is the reported Aero-F nonlinear-ROM timer, including the I/O and postprocessing used to evaluate the C_L, C_D, and C_M integrals. Fully optimized timings without this I/O remain to be measured.

HGV2 Validation Error Comparison

The plotted global RBF uses all 363 ECSW parameter cases with row stride 4.



HGV2 ECSW Offline Cost

ECSW was run on Sherlock using 20 nodes, 480 MPI ranks, and 480 cores. Times are maximum-rank wall times from the offline logs.

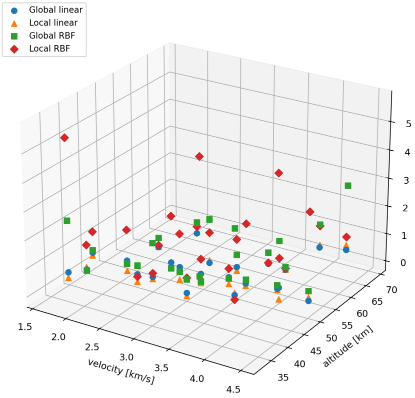
Model	Parameter cases	Rows per case	Row stride	Rows read	Sample nodes	ECSW [h]
Global linear HPROM	30	30^2	1	27,000	**5,089**	30.04
Local linear HPROM, $N_c = 4$	30	30^2	1	27,000	**3,484**	9.95
Global HPROM–RBF, $n = 8$	363	8^2	4	5,808	**1,715**	0.43
Local HPROM–RBF, $N_c = 4$, $n = 8$	30	8^2	1	1,920	**1,113**	0.38

What the ECSW columns mean

Parameter cases $= 30$ means a selected subset of the 363 input configurations. For the linear models, $m_k = \min(r_k, 30) = 30$, hence 30^2 candidate Jacobian rows per selected case. For RBF, $m_k = n = 8$, hence 8^2 rows. Row stride 4 reads rows $0, 4, 8, \ldots$ within every one of the 363 global-RBF cases: $363 \times 8^2/4 = 5{,}808$ rows. It does not subsample parameter cases or online Newton iterations.

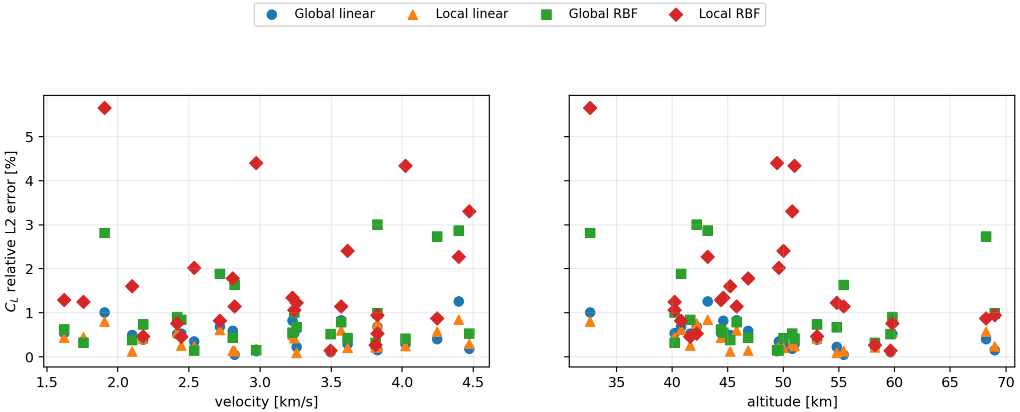
HGV2 Three-Dimensional C_L Error Field

Each marker is one velocity–altitude pair. Its vertical coordinate is the relative L^2 C_L error over the four sampled angles of attack.



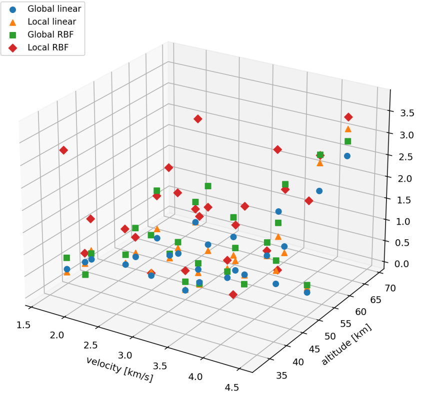
HGV2 Two-Dimensional C_L Error Projections

The front and side projections of the pairwise error field: velocity versus error on the left, altitude versus error on the right.



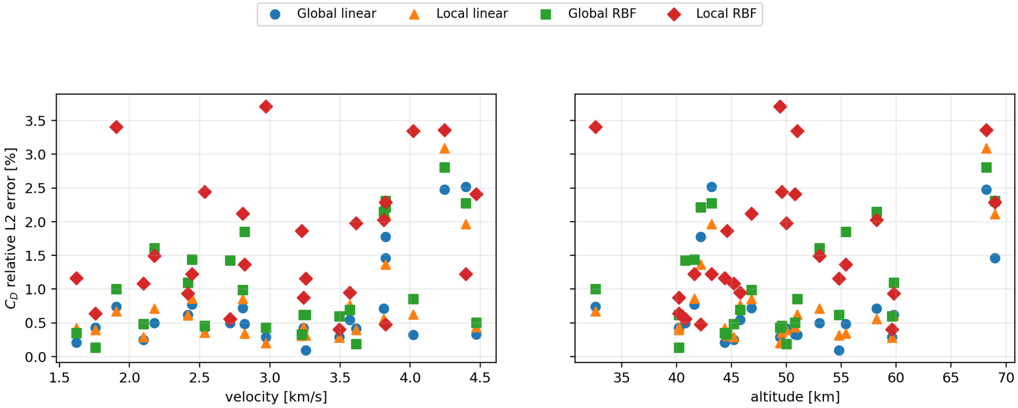
HGV2 Three-Dimensional C_D Error Field

Each marker is one velocity–altitude pair. Its vertical coordinate is the relative L^2 C_D error over the four sampled angles of attack.



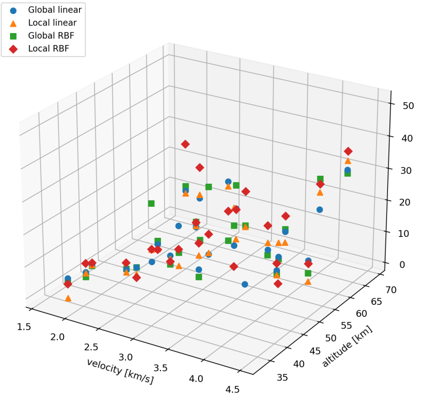
HGV2 Two-Dimensional C_D Error Projections

The front and side projections of the pairwise error field: velocity versus error on the left, altitude versus error on the right.



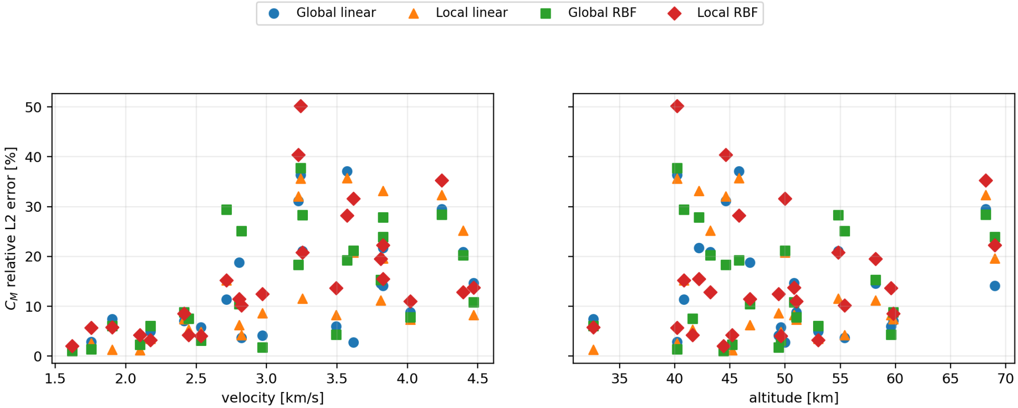
HGV2 Three-Dimensional C_M Error Field

Each marker is one velocity–altitude pair. Its vertical coordinate is the relative L^2 C_M error over the four sampled angles of attack.



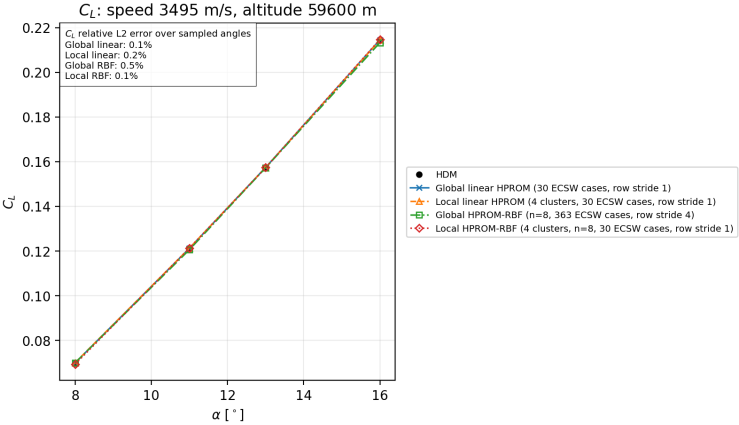
HGV2 Two-Dimensional C_M Error Projections

The front and side projections of the pairwise error field: velocity versus error on the left, altitude versus error on the right.



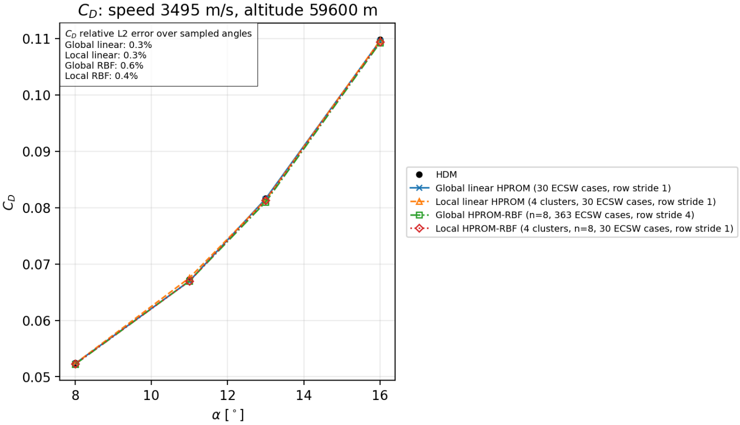
HGV2 Representative Median Pair: C_L

$v = 3.495$ km/s, altitude $= 59.6$ km. This is the median local-RBF pair after ranking the 25 pairs by the combined C_L, C_D, and C_M relative-error norm.



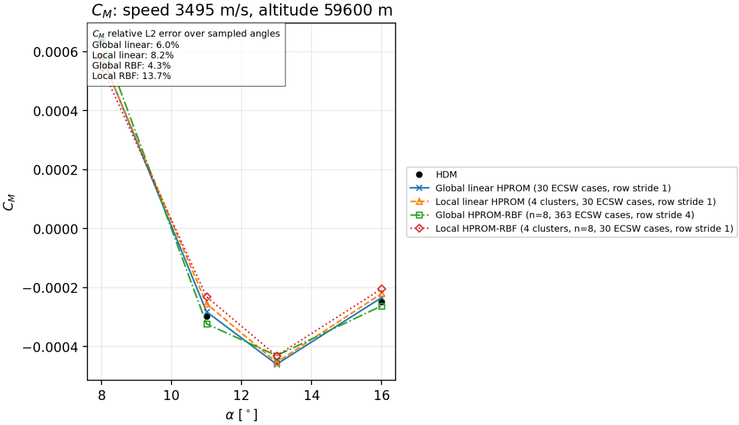
HGV2 Representative Median Pair: C_D

$v = 3.495$ km/s, altitude $= 59.6$ km. This is the median local-RBF pair after ranking the 25 pairs by the combined C_L, C_D, and C_M relative-error norm.



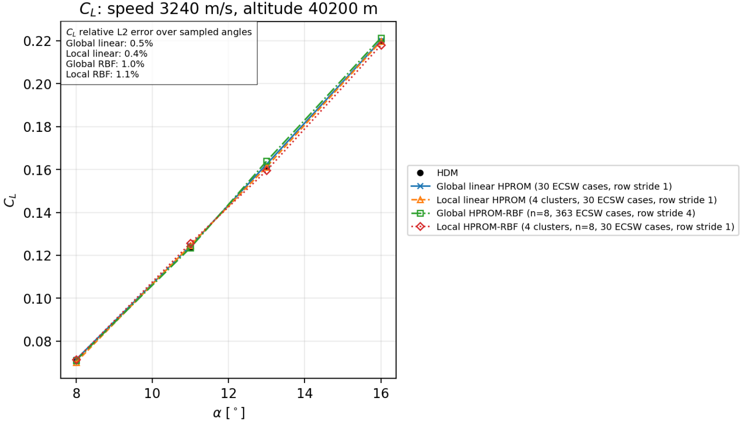
HGV2 Representative Median Pair: C_M

$v = 3.495$ km/s, altitude $= 59.6$ km. This is the median local-RBF pair after ranking the 25 pairs by the combined C_L, C_D, and C_M relative-error norm.



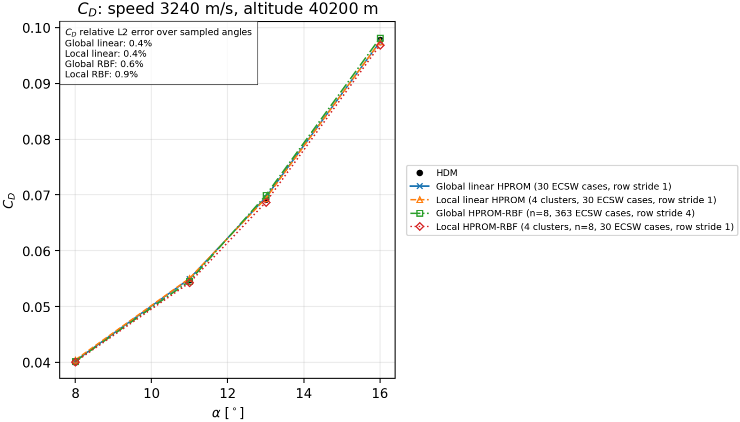
HGV2 Representative Challenging Pair: C_L

$v = 3.240$ km/s, altitude $= 40.2$ km. This pair has the largest combined local-RBF error across C_L, C_D, and C_M.



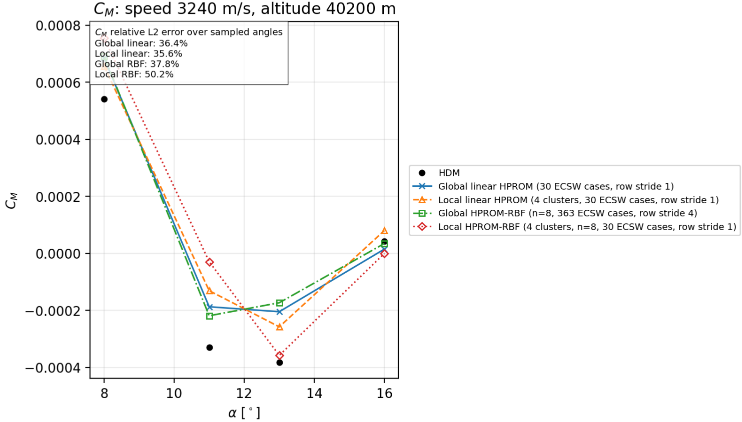
HGV2 Representative Challenging Pair: C_D

$v = 3.240$ km/s, altitude $= 40.2$ km. This pair has the largest combined local-RBF error across C_L, C_D, and C_M.



HGV2 Representative Challenging Pair: C_M

$v = 3.240$ km/s, altitude $= 40.2$ km. This pair has the largest combined local-RBF error across C_L, C_D, and C_M.



HGV2 Conclusions

- Global and local linear HPROM give the best validation accuracy. Global linear is the best in C_D; local linear is the best in C_L and C_M.
- The global RBF with 363 ECSW cases and row stride 4 is stable and reduces the global-RBF moment error relative to the 30-case, stride-1 ablation, but it does not surpass the linear models.
- The local RBF is the fastest online model. Its C_M error is comparable to global RBF, but its C_L and C_D errors are larger. The next controlled test should vary one ECSW factor at a time and broaden the RBF kernel/hyperparameter search.

Appendix

Supplementary results and implementation details

Appendix: ECSW Training Data

For `DataType = Jacobian`, each selected parameter solution assigned to cluster k supplies m_k^2 candidate ECSW rows, where $m_k = \min(r_k,\ \texttt{MaxNumResidualComponents})$ and $\texttt{MaxNumResidualComponents} = 30$. With row stride s, CM reads every s-th row of its `training_rows` sequence.

Model	**ECSW cases**	**Row stride**	m_k	**Rows read by CM**
Global linear HPROM	30	1	$\min(362, 30) = 30$	$30 \times 30^2 = 27{,}000$
Local linear HPROM, $N_c = 4$	30	1	$\min(r_k, 30) = 30$	$30 \times 30^2 = 27{,}000$
Global HPROM–RBF, $n = 8$ (ablation)	30	1	$\min(8, 30) = 8$	$30 \times 8^2 = 1{,}920$
Global HPROM–RBF, $n = 8$ (final)	363	4	$\min(8, 30) = 8$	$363 \times 8^2/4 = 5{,}808$
Local HPROM–RBF, $N_c = 4$, $n = 8$	30	1	$\min(8, 30) = 8$	$30 \times 8^2 = 1{,}920$

Interpretation

An ECSW snapshot is a Jacobian-training row, not a physical-time state. The RBF primary dimension $n = 8$—not $\bar{n}$—sets the 8×8 count. A row stride of 4 reads rows $0, 4, 8, \ldots$ within every selected parameter case; it does not subsample the 363 parameter cases and does not refer to online Newton iterations.

Appendix: HGV2 Reduced-Mesh Construction

Model	L0	L1	L2	L3	Nodes	Elems.	$N_{\neg L2}$
Global linear HPROM	5,089	36,971	166,878	60,623	269,561	563,301	92,683
Local linear HPROM, 4 clusters	3,484	28,145	169,798	47,183	248,610	470,649	78,812
Global HPROM–RBF, $n = 8$	1,715	15,809	173,072	28,648	219,244	350,921	46,172
Local HPROM–RBF, 4 clusters, $n = 8$	1,113	10,516	175,813	20,602	208,044	298,219	32,231

L0: selected ECSW sample nodes.
L1: nodes added by edge connectivity for linear reconstruction.

L2: lift-surface nodes used only for C_L, C_D, and C_M postprocessing.
L3: nodes added by element connectivity for state reconstruction.

$N_{\neg L2}$ excludes the postprocessing-only lift-surface nodes.

Global ROM Storyline: LSPG + Gauss-Newton

Given the implicit HDM residual $\mathbf{r}(\mathbf{u}) = \mathbf{0}$, define a manifold

$$\mathbf{u} \approx \mathbf{u}_{\mathrm{ref}} + \widetilde{\mathbf{u}}(\mathbf{q}).$$

LSPG solves at each time step:

$$\mathbf{q}^{\star} = \arg\min_{\mathbf{q}} \frac{1}{2} \left\| \mathbf{r}(\mathbf{u}_{\mathrm{ref}} + \widetilde{\mathbf{u}}(\mathbf{q})) \right\|_2^2 .$$

With Gauss-Newton (current iterate $\mathbf{q}^{(k)}$), the increment $\Delta\mathbf{q}$ is obtained from

$$\left(\boldsymbol{\Psi}^T \mathbf{J}^T \mathbf{J} \boldsymbol{\Psi}\right) \Delta\mathbf{q} = -\boldsymbol{\Psi}^T \mathbf{J}^T \mathbf{r},$$

where

$$\boldsymbol{\Psi} = \frac{\partial \widetilde{\mathbf{u}}}{\partial \mathbf{q}}, \qquad \mathbf{J} = \frac{\partial \mathbf{r}}{\partial \mathbf{u}}, \qquad \mathbf{q}^{(k+1)} = \mathbf{q}^{(k)} + \Delta\mathbf{q}.$$

Global Linear HPROM

State ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\mathrm{ref}} + \mathbf{V}\mathbf{q}, \qquad \mathbf{V} \in \mathbb{R}^{N \times n}, \; n = 96.$$

Hence

$$\widetilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}\mathbf{q}, \qquad \boldsymbol{\Psi} = \frac{\partial \widetilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V}.$$

Gauss-Newton system:

$$\left(\mathbf{V}^T \mathbf{J}^T \mathbf{J} \mathbf{V}\right) \Delta \mathbf{q} = -\mathbf{V}^T \mathbf{J}^T \mathbf{r}.$$

Global HQPROM

Quadratic-manifold ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\mathrm{ref}} + \mathbf{V}\mathbf{q} + \mathbf{H}\left(\mathbf{q} \otimes \mathbf{q}\right), \qquad n = 39.$$

Thus

$$\widetilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}\mathbf{q} + \mathbf{H}\left(\mathbf{q} \otimes \mathbf{q}\right),$$

and the tangent depends on the current reduced state:

$$\boldsymbol{\Psi}(\mathbf{q}) = \frac{\partial \widetilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V} + \frac{\partial}{\partial \mathbf{q}}\left[\mathbf{H}\left(\mathbf{q} \otimes \mathbf{q}\right)\right].$$

Global HPROM-GPR

Split basis and reduced coordinates:

$$\mathbf{V}_{\mathrm{tot}} = [\mathbf{V}, \bar{\mathbf{V}}], \qquad \mathbf{q}_{\mathrm{tot}} = \begin{bmatrix} \mathbf{q} \\ \bar{\mathbf{q}} \end{bmatrix}.$$

$$\bar{\mathbf{q}} = \mathcal{N}_{\mathrm{GPR}}(\mathbf{q}), \qquad n = 20, \ \bar{n} = 131.$$

State ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\mathrm{ref}} + \widetilde{\mathbf{u}}(\mathbf{q}),$$

$$\widetilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}_{\mathrm{tot}}\mathbf{q}_{\mathrm{tot}} = \mathbf{V}\mathbf{q} + \bar{\mathbf{V}}\,\mathcal{N}_{\mathrm{GPR}}(\mathbf{q}).$$

Effective tangent for Gauss-Newton:

$$\boldsymbol{\Psi}(\mathbf{q}) = \frac{\partial \widetilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V} + \bar{\mathbf{V}}\frac{\partial \mathcal{N}_{\mathrm{GPR}}}{\partial \mathbf{q}}.$$

Local ROMs: Piecewise Manifolds

Partition the training snapshots into clusters $k = 1, \ldots, N_c$ and build one local trial manifold per cluster:

$$\mathbf{u} \approx \Phi_k(\mathbf{q}_k).$$

Examples used in this work:

$$\Phi_k^{\mathrm{lin}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k \mathbf{q},$$

$$\Phi_k^{\mathrm{quad}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k \mathbf{q} + \mathbf{H}_k(\mathbf{q} \otimes \mathbf{q}),$$

$$\Phi_k^{\mathrm{gpr}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k \mathbf{q} + \bar{\mathbf{V}}_k \mathcal{N}_{\mathrm{GPR},k}(\mathbf{q}).$$

Online workflow

1. Select the active cluster.
2. If the cluster changes, transfer the current state to the new local manifold.
3. Solve the local LSPG problem in the active manifold.

Cluster Selection: Final Form (Linear)

Shared nearest-centroid selector (used in all local models):

$$k^+ = \arg\min_{\ell=1,\ldots,N_c} \|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2, \qquad \mathbf{u}^s = \Phi_{k^-}(\mathbf{q}_{k^-}).$$

For the linear local manifold $\Phi^{\mathrm{lin}}_{k^-}(\mathbf{q}) = \mathbf{u}_{0,k^-} + \mathbf{V}_{k^-}\mathbf{q}$:

$$k^+ = \arg\min_{\ell=1,\ldots,N_c} S^{\mathrm{lin}}_{k^-,\ell}(\mathbf{q}_{k^-}),$$

$$S^{\mathrm{lin}}_{k^-,\ell}(\mathbf{q}) = 2\,\mathbf{g}^T_{k^-,\ell}\mathbf{q} + d_{k^-,\ell},$$

with offline-precomputed terms

$$\mathbf{g}_{k^-,\ell} := \mathbf{V}^T_{k^-}(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}), \qquad d_{k^-,\ell} := \|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2.$$

Cluster Selection: Final Form (HQPROM and HPROM-GPR)

HQPROM selector

$$k^{+} = \arg\min_{\ell} S^{\mathrm{quad}}_{k^-,\ell}(\mathbf{q}_{k^-}),$$

$$S^{\mathrm{quad}}_{k^-,\ell}(\mathbf{q}) = 2\,\mathbf{g}^T_{k^-,\ell}\mathbf{q} + 2\,\mathbf{m}^T_{k^-,\ell}(\mathbf{q}\otimes\mathbf{q}) + d_{k^-,\ell},$$

$$\mathbf{m}^T_{k^-,\ell}(\mathbf{q}\otimes\mathbf{q}) := (\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell})^T\mathbf{H}_{k^-}(\mathbf{q}\otimes\mathbf{q}).$$

HPROM-GPR selector

$$k^{+} = \arg\min_{\ell} S^{\mathrm{gpr}}_{k^-,\ell}(\mathbf{q}_{k^-}),$$

$$S^{\mathrm{gpr}}_{k^-,\ell}(\mathbf{q}) = 2\,\mathbf{g}^T_{k^-,\ell}\mathbf{q} + 2\,\bar{\mathbf{g}}^T_{k^-,\ell}\mathcal{N}_{\mathrm{GPR},k^-}(\mathbf{q}) + d_{k^-,\ell}.$$

$$\bar{\mathbf{q}}_{k^-} := \mathcal{N}_{\mathrm{GPR},k^-}(\mathbf{q}_{k^-}), \qquad \bar{\mathbf{g}}_{k^-,\ell} := \bar{\mathbf{V}}^T_{k^-}(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}).$$

Cluster Change: Linear and Nonlinear Transfers

$$k^+ = k^- \Rightarrow \text{no transfer}, \qquad k^+ \neq k^- \Rightarrow \text{solve transfer problem}.$$

Transferred state:

$$\mathbf{u}^s = \Phi_{k^-}(\mathbf{q}_{k^-}).$$

Linear local target

$$\mathbf{q}_{k^+}^{\mathrm{lin}} = \arg\min_{\mathbf{q}} \left\| \mathbf{u}^s - \left(\mathbf{u}_{0,k^+} + \mathbf{V}_{k^+}\mathbf{q}\right) \right\|_2^2,$$

$$\text{if } \mathbf{V}_{k^+}^T \mathbf{V}_{k^+} = \mathbf{I}, \quad \mathbf{q}_{k^+}^{\mathrm{lin}} = \mathbf{V}_{k^+}^T(\mathbf{u}^s - \mathbf{u}_{0,k^+}).$$

Nonlinear local targets (HQPROM / HPROM-GPR)

$$\mathbf{q}_{k^+}^{\mathrm{nl}} = \arg\min_{\mathbf{q}} \left\| \mathbf{u}^s - \Phi_{k^+}^{\mathrm{nl}}(\mathbf{q}) \right\|_2^2.$$

Cluster Change: Nonlinear Least Squares

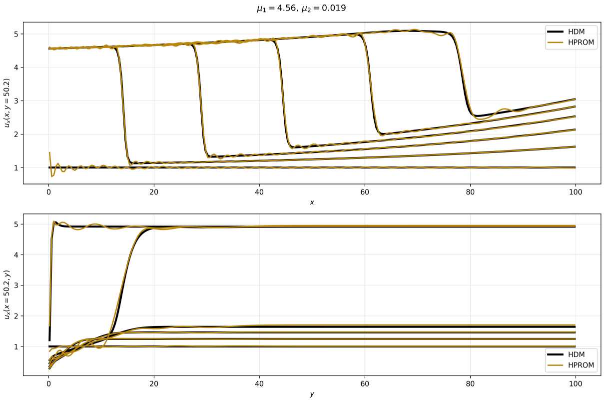
For nonlinear local manifolds, define

$$\mathbf{r}_{\mathrm{tr}}(\mathbf{q}) := \Phi_{k^+}(\mathbf{q}) - \mathbf{u}^s, \qquad \mathbf{J}_{\Phi}(\mathbf{q}) := \frac{\partial \Phi_{k^+}}{\partial \mathbf{q}}.$$

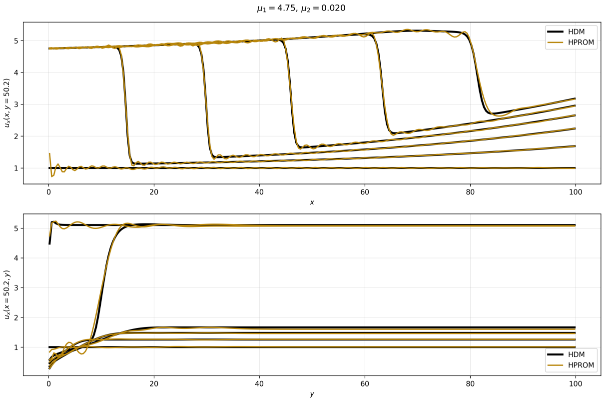
Gauss–Newton at iterate $\mathbf{q}^{(j)}$:

$$\left(\mathbf{J}_{\Phi}^T \mathbf{J}_{\Phi}\right) \Delta\mathbf{q} = -\mathbf{J}_{\Phi}^T \mathbf{r}_{\mathrm{tr}}(\mathbf{q}^{(j)}), \qquad \mathbf{q}^{(j+1)} = \mathbf{q}^{(j)} + \Delta\mathbf{q}.$$

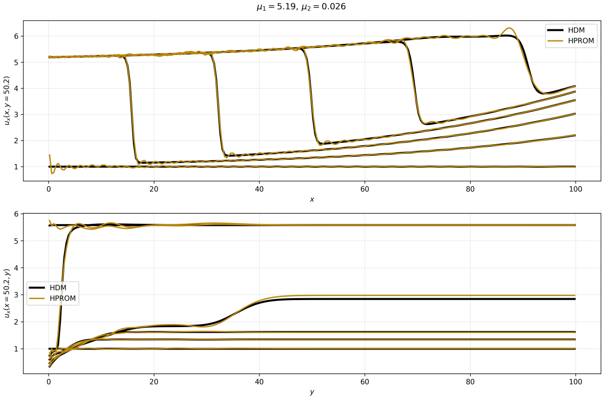
HPROM vs HDM: $\mu = (4.56, 0.019)$



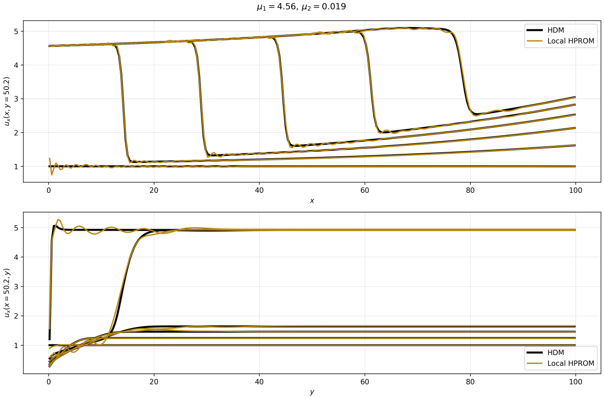
HPROM vs HDM: $\boldsymbol{\mu} = (4.75, 0.020)$



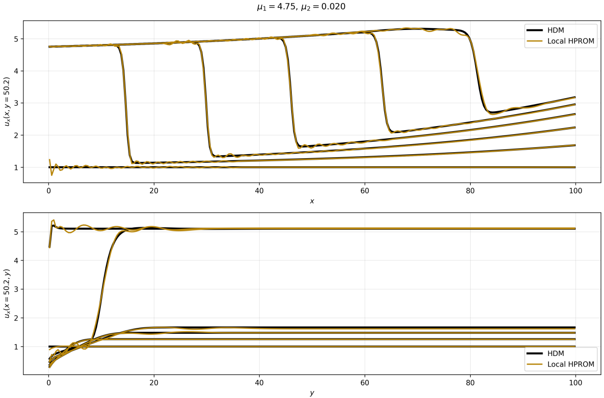
HPROM vs HDM: $\boldsymbol{\mu} = (5.19, 0.026)$



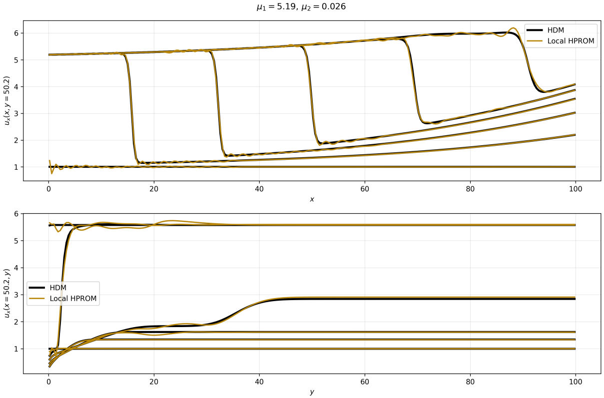
Local HPROM ($N_c = 3$) vs HDM: $\boldsymbol{\mu} = (4.56, 0.019)$



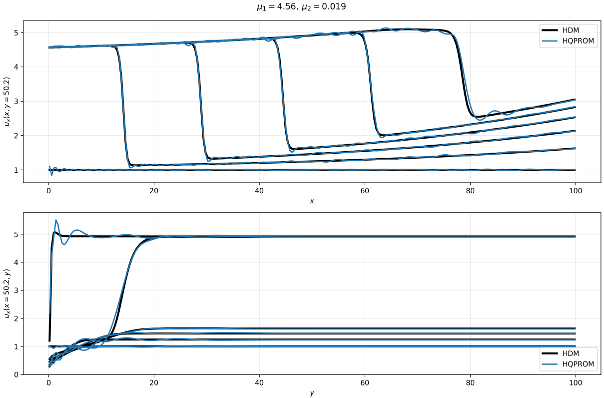
Local HPROM ($N_c = 3$) vs HDM: $\boldsymbol{\mu} = (4.75, 0.020)$



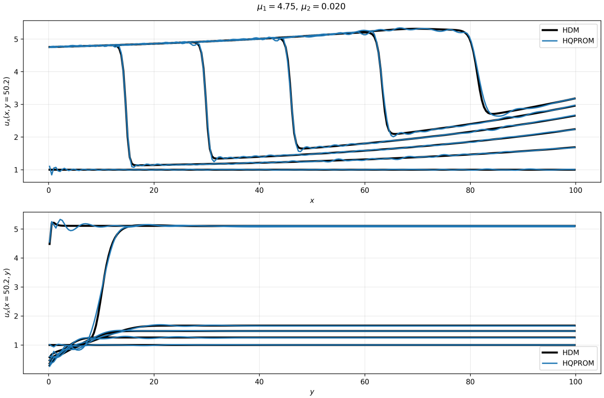
Local HPROM ($N_c = 3$) vs HDM: $\boldsymbol{\mu} = (5.19, 0.026)$



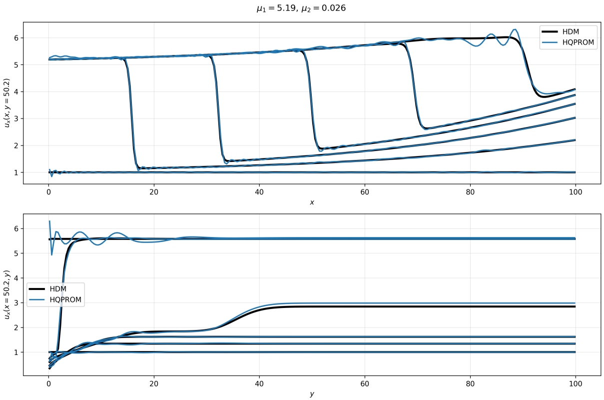
HQPROM vs HDM: $\boldsymbol{\mu} = (4.56,\ 0.019)$



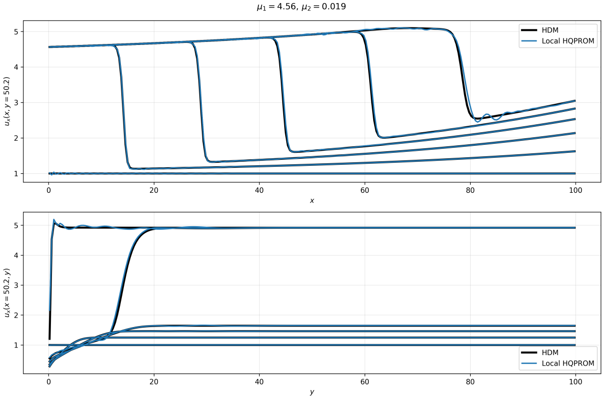
HQPROM vs HDM: $\boldsymbol{\mu} = (4.75,\ 0.020)$



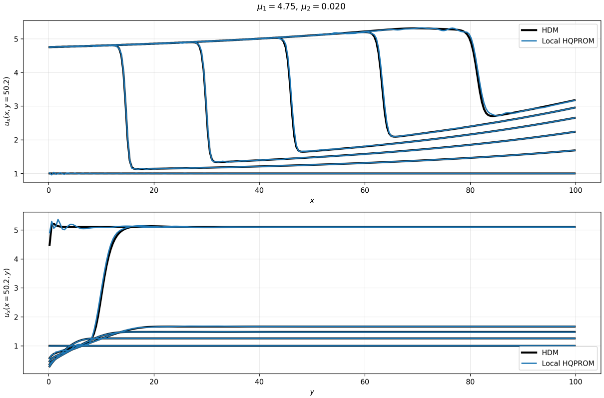
HQPROM vs HDM: $\boldsymbol{\mu} = (5.19,\ 0.026)$



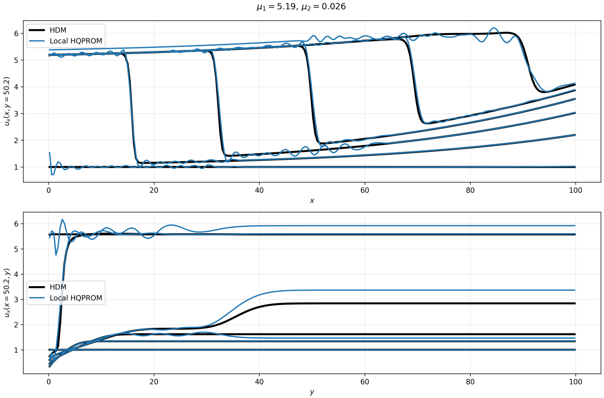
Local HQPROM ($N_c = 3$) vs HDM: $\boldsymbol{\mu} = (4.56, 0.019)$



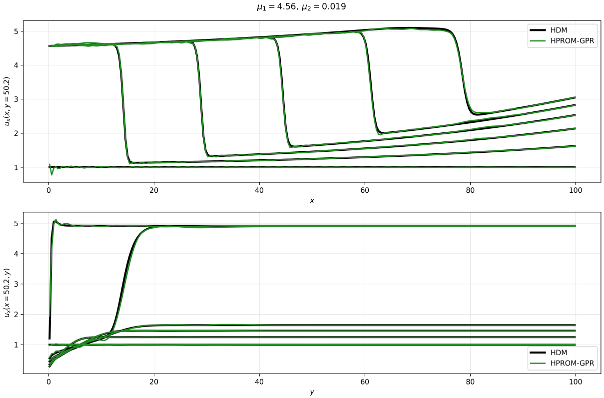
Local HQPROM ($N_c = 3$) vs HDM: $\boldsymbol{\mu} = (4.75, 0.020)$



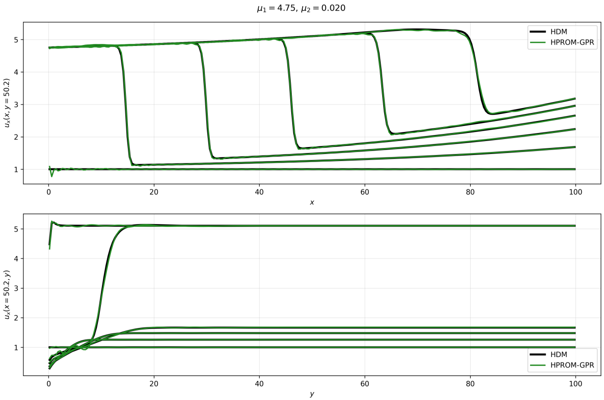
Local HQPROM ($N_c = 3$) vs HDM: $\boldsymbol{\mu} = (5.19, 0.026)$



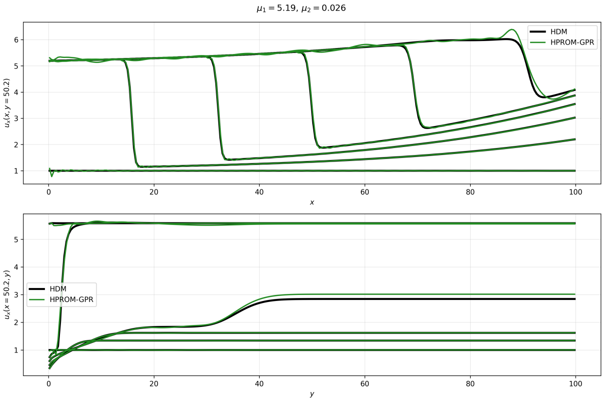
HPROM-GPR vs HDM: $\boldsymbol{\mu} = (4.56, 0.019)$



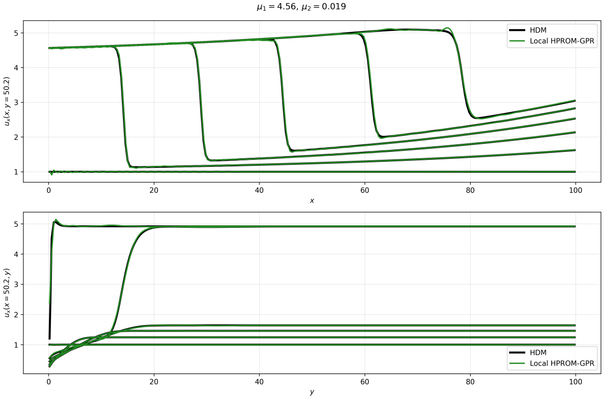
HPROM-GPR vs HDM: $\boldsymbol{\mu} = (4.75,\ 0.020)$



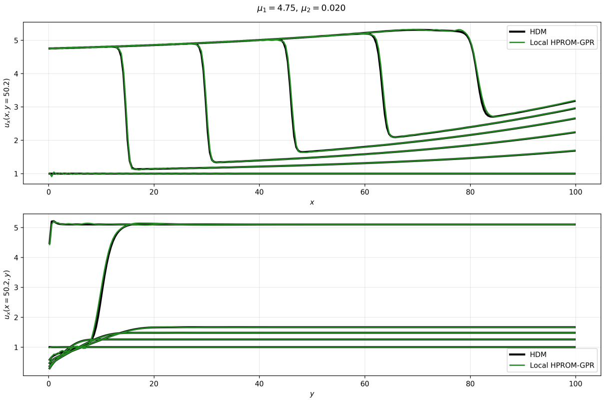
HPROM-GPR vs HDM: $\boldsymbol{\mu} = (5.19,\ 0.026)$



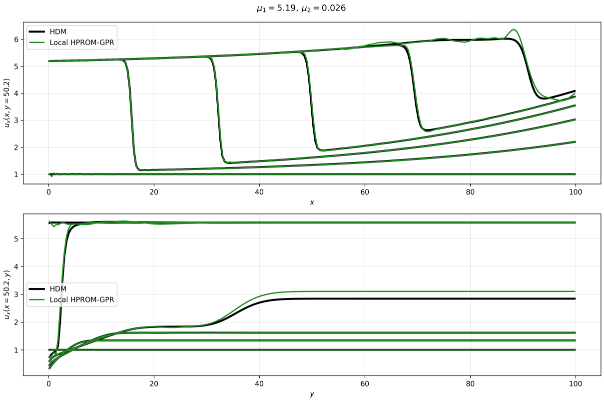
Local HPROM-GPR ($N_c = 3$) vs HDM: $\boldsymbol{\mu} = (4.56, 0.019)$



Local HPROM-GPR ($N_c = 3$) vs HDM: $\boldsymbol{\mu} = (4.75, 0.020)$



Local HPROM-GPR ($N_c = 3$) vs HDM: $\boldsymbol{\mu} = (5.19, 0.026)$



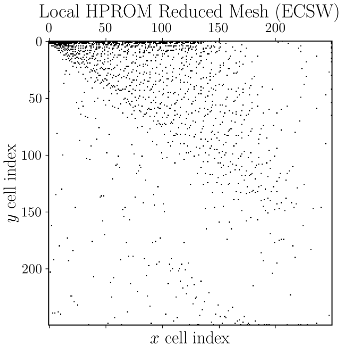
Appendix: Local ROM Results at $N_c = 10$

Computational model	n	$\bar{n}$	N_e	avg $\mathbb{RE}_{2,\mathbf{u}}$ (%)	max $\mathbb{RE}_{2,\mathbf{u}}$ (%)	Wall-clock time (min)	Speedup
Local HPROM ($N_c = 10$)	14–23	–	1 522	0.867	0.956	0.198	63.257
Local HQPROM ($N_c = 10$)	8–12	–	871	0.572	0.929	0.263	47.497
Local HPROM-GPR ($N_c = 10$)	6	24–47	541	0.604	0.801	0.244	51.261

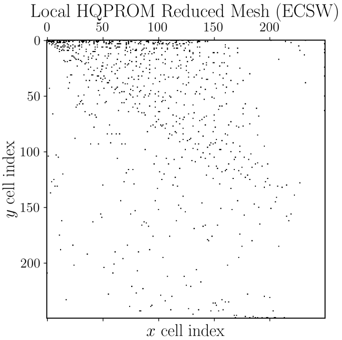
Speedup is computed with respect to the HDM runtime, $t_{\mathrm{HDM}} = 750.702$ s.

Global-model rows are unaffected by N_c and are reproduced in the main results table.

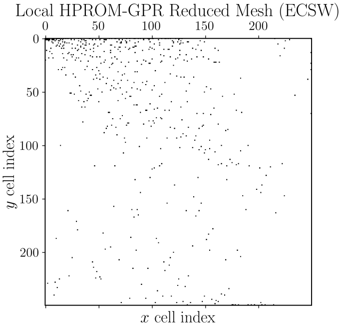
Appendix: Local Reduced Meshes (ECSW), $N_c = 10$



Local HPROM (N_e = 1,522)

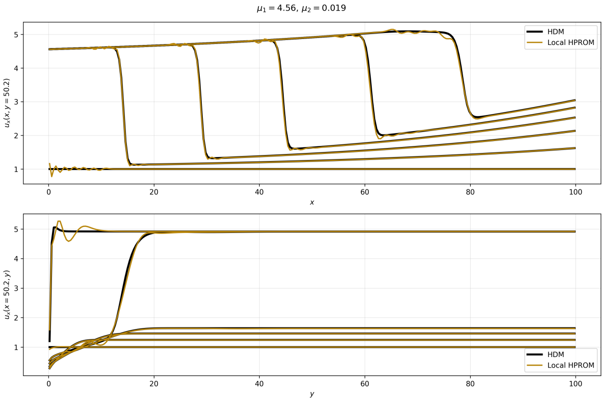


Local HQPROM (N_e = 871)

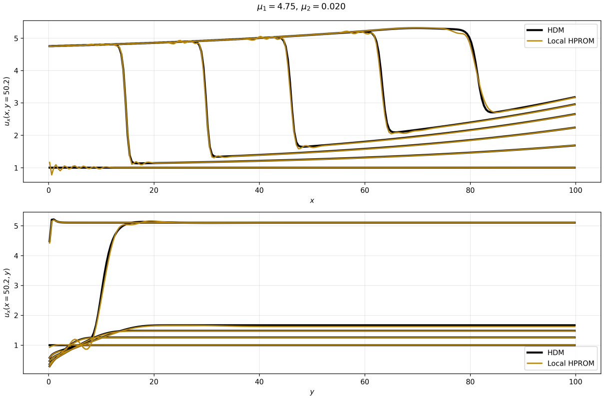


Local HPROM-GPR (N_e = 541)

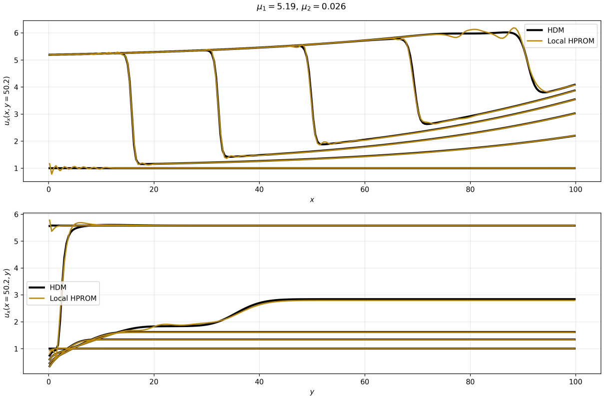
Appendix: Local HPROM ($N_c = 10$) vs HDM: $\boldsymbol{\mu} = (4.56, 0.019)$



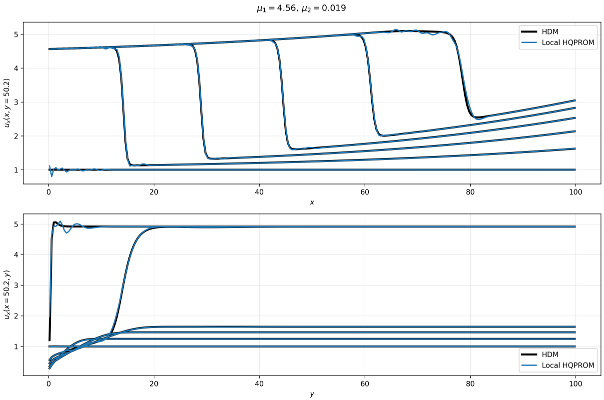
Appendix: Local HPROM ($N_c = 10$) vs HDM: $\boldsymbol{\mu} = (4.75, 0.020)$



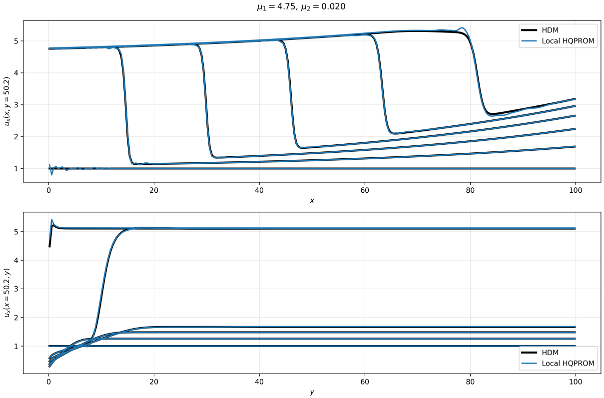
Appendix: Local HPROM ($N_c = 10$) vs HDM: $\boldsymbol{\mu} = (5.19, 0.026)$



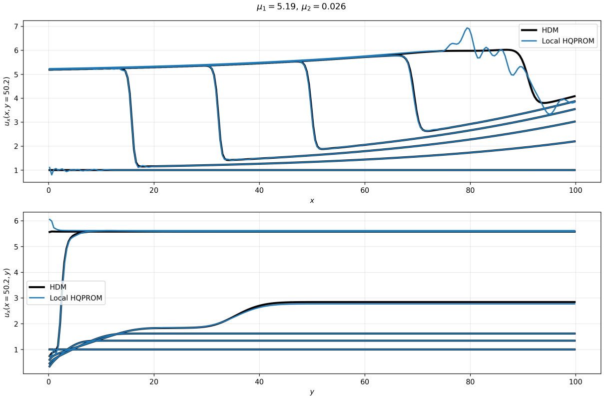
Appendix: Local HQPROM ($N_c = 10$) vs HDM: $\boldsymbol{\mu} = (4.56, 0.019)$



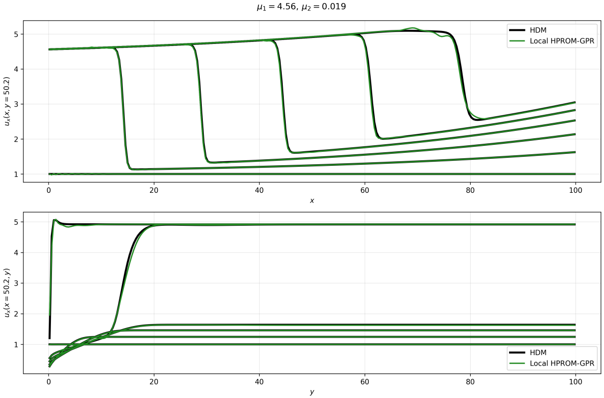
Appendix: Local HQPROM ($N_c = 10$) vs HDM: $\boldsymbol{\mu} = (4.75, 0.020)$



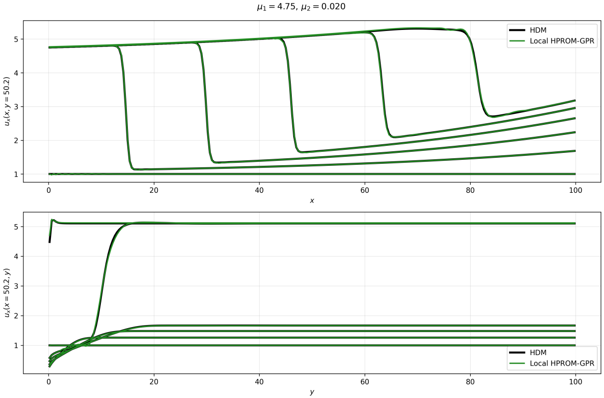
Appendix: Local HQPROM ($N_c = 10$) vs HDM: $\boldsymbol{\mu} = (5.19, 0.026)$



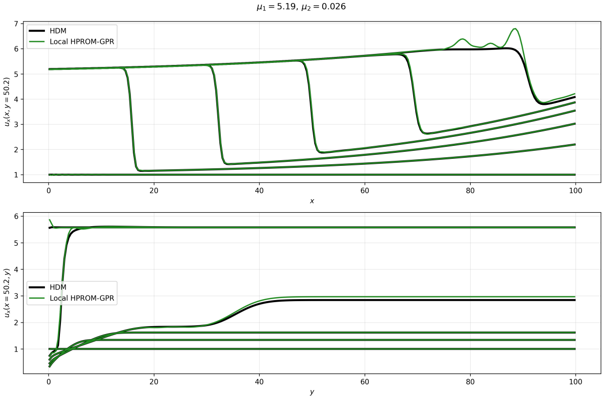
Appendix: Local HPROM-GPR ($N_c = 10$) vs HDM: $\boldsymbol{\mu} = (4.56, 0.019)$



Appendix: Local HPROM-GPR ($N_c = 10$) vs HDM: $\boldsymbol{\mu} = (4.75, 0.020)$



Appendix: Local HPROM-GPR ($N_c = 10$) vs HDM: $\boldsymbol{\mu} = (5.19, 0.026)$



Appendix: Linear Selector Expansion

$$\mathbf{u}^s = \mathbf{u}_{0,k^-} + \mathbf{V}_{k^-}\mathbf{q}_{k^-}, \qquad k^+ = \arg\min_{\ell} \|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2.$$

$$\text{Let } \mathbf{a}_\ell := \mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}, \quad \mathbf{b} := \mathbf{V}_{k^-}\mathbf{q}_{k^-}.$$

$$\|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2 = \|\mathbf{a}_\ell + \mathbf{b}\|_2^2 = \|\mathbf{a}_\ell\|_2^2 + \|\mathbf{b}\|_2^2 + 2\,\mathbf{a}_\ell^T\mathbf{b}.$$

$$\|\mathbf{b}\|_2^2 = \|\mathbf{V}_{k^-}\mathbf{q}_{k^-}\|_2^2 \text{ is independent of } \ell.$$

$$\Rightarrow \; k^+ = \arg\min_{\ell} \left[\|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2 + 2(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell})^T\mathbf{V}_{k^-}\mathbf{q}_{k^-} \right].$$

Precompute

$$\mathbf{g}_{k^-,\ell} := \mathbf{V}_{k^-}^T(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}), \qquad d_{k^-,\ell} := \|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2,$$

which yields

$$k^+ = \arg\min_{\ell} \left(2\,\mathbf{g}_{k^-,\ell}^T\mathbf{q}_{k^-} + d_{k^-,\ell} \right).$$

Appendix: HQPROM Selector Expansion

$$\mathbf{u}^s = \mathbf{u}_{0,k-} + \mathbf{V}_{k-}\mathbf{q}_{k-} + \mathbf{H}_{k-}(\mathbf{q}_{k-} \otimes \mathbf{q}_{k-}).$$

$$\text{Let } \mathbf{a}_\ell := \mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}, \quad \mathbf{b} := \mathbf{V}_{k-}\mathbf{q}_{k-}, \quad \mathbf{c} := \mathbf{H}_{k-}(\mathbf{q}_{k-} \otimes \mathbf{q}_{k-}).$$

$$k^+ = \arg\min_\ell \|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2 = \arg\min_\ell \|\mathbf{a}_\ell + \mathbf{b} + \mathbf{c}\|_2^2.$$

$$\|\mathbf{a}_\ell + \mathbf{b} + \mathbf{c}\|_2^2 = \|\mathbf{a}_\ell\|_2^2 + \|\mathbf{b}\|_2^2 + \|\mathbf{c}\|_2^2 + 2\,\mathbf{a}_\ell^T\mathbf{b} + 2\,\mathbf{a}_\ell^T\mathbf{c} + 2\,\mathbf{b}^T\mathbf{c}.$$

$$\|\mathbf{b}\|_2^2,\ \|\mathbf{c}\|_2^2,\ 2\,\mathbf{b}^T\mathbf{c} \text{ are independent of } \ell.$$

$$\Rightarrow\ k^+ = \arg\min_\ell \Big[\|\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}\|_2^2$$

$$+2(\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell})^T\mathbf{V}_{k-}\mathbf{q}_{k-} + 2(\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell})^T\mathbf{H}_{k-}(\mathbf{q}_{k-} \otimes \mathbf{q}_{k-})\Big].$$

Define

$$\mathbf{m}_{k-,\ell}^T(\mathbf{q} \otimes \mathbf{q}) := (\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell})^T\mathbf{H}_{k-}(\mathbf{q} \otimes \mathbf{q}),$$

then

$$k^+ = \arg\min_\ell \left(2\,\mathbf{g}_{k-,\ell}^T\mathbf{q}_{k-} + 2\,\mathbf{m}_{k-,\ell}^T(\mathbf{q}_{k-} \otimes \mathbf{q}_{k-}) + d_{k-,\ell}\right).$$

Appendix: HPROM-GPR Selector Expansion

$$\mathbf{u}^s = \mathbf{u}_{0,k-} + \mathbf{V}_{k-}\mathbf{q}_{k-} + \bar{\mathbf{V}}_{k-}\bar{\mathbf{q}}_{k-}, \qquad \bar{\mathbf{q}}_{k-} = \mathcal{N}_{\mathrm{GPR},k-}(\mathbf{q}_{k-}).$$

$$\text{Let } \mathbf{a}_\ell := \mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}, \quad \mathbf{b} := \mathbf{V}_{k-}\mathbf{q}_{k-}, \quad \mathbf{c} := \bar{\mathbf{V}}_{k-}\bar{\mathbf{q}}_{k-}.$$

$$\|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2 = \|\mathbf{a}_\ell + \mathbf{b} + \mathbf{c}\|_2^2$$

$$= \|\mathbf{a}_\ell\|_2^2 + \|\mathbf{b}\|_2^2 + \|\mathbf{c}\|_2^2 + 2\,\mathbf{a}_\ell^T\mathbf{b} + 2\,\mathbf{a}_\ell^T\mathbf{c} + 2\,\mathbf{b}^T\mathbf{c}.$$

$$\|\mathbf{b}\|_2^2,\ \|\mathbf{c}\|_2^2,\ 2\,\mathbf{b}^T\mathbf{c} \text{ are independent of } \ell.$$

$$\Rightarrow\ k^+ = \arg\min_{\ell}\left[\|\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}\|_2^2 + 2(\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell})^T\mathbf{V}_{k-}\mathbf{q}_{k-} + 2(\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell})^T\bar{\mathbf{V}}_{k-}\bar{\mathbf{q}}_{k-}\right].$$

Hence, with

$$\bar{\mathbf{g}}_{k-,\ell} := \bar{\mathbf{V}}_{k-}^T(\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}),$$

$$k^+ = \arg\min_{\ell}\left(2\,\mathbf{g}_{k-,\ell}^T\mathbf{q}_{k-} + 2\,\bar{\mathbf{g}}_{k-,\ell}^T\bar{\mathbf{q}}_{k-} + d_{k-,\ell}\right).$$